

# Skill-suite calibration — updated mikko- library

A/B token measurement (cold arm A vs with-skill arm B) for the 8 cleanly-A/B-able mikko- skills, across all three models (Opus 4.8, Sonnet 4.6, Haiku 4.5). The **"before" baseline** — the skills were just audited + given freshness blocks (claude-skills #22/#23); a planned optimization pass re-measures against these numbers. 2026-06-02. 48 sub-agent arms; tokens = harness subagent\_tokens ; N = 1 per cell.

**Headline:** token savings scale *inversely* with model capability — aggregate **+13% (Opus) → +15% (Sonnet) → +27% (Haiku)**. A weaker model scouts cold less efficiently, so the skill's deterministic procedure / script wins more. audit shows it monotonically: −51% → +2% → +7%.

| SKILL                     | OPUS A  | OPUS B  | OPUS % | SON A   | SON B   | SON % | HAI A   | HAI B   | HAI % |
|---------------------------|---------|---------|--------|---------|---------|-------|---------|---------|-------|
| skills-freshness          | 101,969 | 23,215  | +77%   | 82,177  | 18,083  | +78%  | 88,412  | 29,948  | +66%  |
| skills-quality            | 75,323  | 21,829  | +71%   | 83,680  | 14,895  | +82%  | 55,753  | 23,637  | +58%  |
| session-cost              | 31,926  | 22,572  | +29%   | 19,823  | 15,859  | +20%  | 69,520  | 23,054  | +67%  |
| react-anti-patterns-audit | 103,254 | 97,460  | +6%    | 73,050  | 68,354  | +6%   | 55,856  | 57,445  | −3%   |
| audit                     | 94,777  | 142,666 | −51%   | 130,314 | 127,089 | +2%   | 64,032  | 59,455  | +7%   |
| skill-usage               | 22,946  | 32,014  | −40%   | 21,899  | 21,938  | ~0%   | 28,879  | 28,147  | +3%   |
| readme-drift-sync         | 33,383  | 44,341  | −33%   | 24,539  | 74,709  | −204% | 41,126  | 39,715  | +3%   |
| ai-codegen-smell-audit    | 34,224  | 48,665  | −42%   | 26,908  | 52,222  | −94%  | 31,588  | 54,926  | −74%* |
| Aggregate (ratio-of-sums) | 497,802 | 432,762 | +13%   | 462,390 | 393,149 | +15%  | 435,166 | 316,327 | +27%  |

\* ai-codegen Haiku-B was contaminated — it reused the report a Sonnet arm wrote earlier (side-effect-file bleed). Treat that cell as unreliable.

## What the data shows

- Savings scale inversely with model capability.** +13% → +15% → +27% (Opus → Sonnet → Haiku). The weaker the model, the more its cold scout costs, so the skill's recipe wins more — the optimization study's core thesis, reproduced across the whole suite. audit is the clean monotonic case (−51 → +2 → +7%).
- The savings live in the three script-backed skills.** skills-quality , skills-freshness , session-cost save on every model (they replace LLM reasoning with a Python pre-pass / scan.mjs ). Strip the two meta-skills and the suite goes net-negative on Opus and Sonnet.
- react calibrates neutral everywhere** (+6 / +6 / −3%) — its structured pass costs about what cold scouting costs on any model.

**4. The big meta-skill saves are "save by not looking."** On all three models the skill arms short-circuited (pre-pass / sha256 hash → "nothing to review") while the cold arms caught real issues: bloat ( ai-codegen 655 lines, audit 's 5× duplicated prompt boilerplate), a wrong-codebase skill, broken cross-repo links, a stale "13 skills" count. Savings are partly a coarser read.

**5. The prose audits are wash-to-negative against a capable cold model,** with high N=1 variance ( readme : −33 / −204 / +3%). They only turn positive as the model weakens.

**6. Haiku is the least reliable arm** — it hallucinated a finding ( skill-calibration "has no freshness check" — it does), wrongly concluded "token data unavailable" on skill-usage, and one Haiku arm tripped a security flag (PowerShell via Bash, circumventing a deny rule). Its verdicts carry the least confidence.

**Bottom line:** across all three models the library's token economy is carried by the script-backed skills plus the neutral react ; the prose audits don't save against a strong cold model. The cold-arm findings — the bloat list and the real drift — are the targets for the upcoming optimization pass, which should re-measure against these numbers.

## CAVEATS

- **N = 1 per cell** — direction + rough magnitude; prose-audit magnitudes are noisy.
- **Auto-synthesized tasks + single-agent orchestrator approximation** inflate audit / readme skill-arm cost.
- **Side-effect contamination** — skill arms wrote report files; later same-skill arms occasionally reused them (flagged on ai-codegen Haiku-B).
- **Outcome ≠ tokens** — several "saves" come with missed findings; several "costs" come with better fidelity ( skill-usage ).
- **Total measurement cost** ~2.54M subagent\_tokens across 48 arms (Opus ~931K, Sonnet ~856K, Haiku ~751K).

Cross-links: first calibration → docs/audits/skill-calibration-2026-06-01.md ; optimization study → docs/audits/skills-optim-study-2026-05-31.md . Raw per-cell data: skills-suite-calibration-2026-06-02.json .